

$$50 \text{ AB} \quad \bar{x} = 40 \quad s = 5$$

① 95 %

$$\alpha = 0.05, \quad \frac{\alpha}{2} = 0.025$$

$$Z_{0.025} = 1 - 0.025$$

$$= \Phi(0.975)$$

$$= 1.96$$

$$\bar{x} \pm Z_{\alpha/2} \times \frac{s}{\sqrt{n}} = 40 \pm 1.96 \times \frac{5}{\sqrt{50}} \quad \text{0.708}$$

$$= (38.6123, 41.3877) \#$$

② 99 %

$$\alpha = 0.01, \quad \alpha/2 = 0.005$$

$$Z_{0.005} = 1 - 0.005$$

$$= \Phi(0.995)$$

$$1.8267$$

$$= 2.58$$

$$40 \pm 2.58 \times \frac{5}{\sqrt{50}} = (38.1733, 41.8267) \#$$

③ 72 %

$$\alpha = 0.28 \quad \alpha/2 = 0.14$$

$$Z_{0.14} = 1 - 0.14 = \Phi(0.86)$$

$$= 1.08$$

$$40 \pm 1.08 \times \frac{5}{\sqrt{50}} = (39.2354, 40.7647) \#$$

$$\textcircled{1} \quad \alpha \rightarrow 1 - \dots \%$$

$$\textcircled{2} \quad \alpha/2 \rightarrow \downarrow /2$$

$$\textcircled{3} \quad 1 - (\alpha/2) = \Phi(\dots) = \dots$$

$$\textcircled{4} \quad \bar{x} \pm \underbrace{\quad}_{\downarrow} \times (gtd / \sqrt{\text{sample}})$$